# Changjia (Eric) Yang

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University of Maryland, Baltimore County
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## Research Interests

Computer Vision; Agentic AI; Interactive Visualization; AI Ethics.

## Education

**University of Maryland, Baltimore County** 2025–Present
*Ph.D. in Computer Science, in progress* Baltimore, MD
Advisor: Dr. Rebecca Williams

**The George Washington University** May 2023
*M.S. in Computer Science* Washington, DC

**Hofstra University** December 2019
*M.S. in Business Statistics* Hempstead, NY

**Chongqing Jiaotong University** June 2017
*B.A. in Advertising* Chongqing, China

## Research Experience

**CounterLens: AI/ML Ethics Education** 2025–Present
*Advisor: Dr. Rebecca Williams*

- Study how students use interactive experiments to reason about model behavior, fairness, and trade-offs between evaluation criteria.
- Designed and built the platform's frontend and backend, allowing students to explore counterfactuals, adjust thresholds, and compare demographic groups without writing code.
- Develop classroom experiments using institutional data for a Fall 2026 study of students' model-evaluation skills and ethical reasoning.

**Investigating Contact-Induced Feature Mixing in 3D Shape Correspondence** 2026–Present

- Investigate whether spatial feature aggregation mixes anatomically distinct surfaces near self-contact, contributing to left–right correspondence errors on 3D human shapes.
- Plan a controlled comparison of GeoLoRA's original aggregation and a surface-aware variant, keeping input shapes, rendered views, and extracted image features fixed.
- Plan evaluation on FAUST across separated, nearby, and contacting body parts, measuring left–right and overall correspondence errors by subject while accounting for visibility, pose, and prediction coverage.

**V-Audit: Visual Evidence Selection for Detecting Collateral Errors in GUI Agents** 2026–Present

- Investigate whether visual evidence selection under a limited review budget helps users detect unintended GUI-agent edits without increasing rejection of correct outputs.
- Propose a method for selecting before-and-after comparisons based on requested edits and content that should remain unchanged, showing both task completion and possible unintended changes.
- Plan an evaluation using audited agent outputs and a controlled user study, comparing against full inspection, difference views, and requirement-linked evidence; measure missed errors, false rejection, review time, and confidence calibration.

**Do Multimodal LLMs Understand Misleading Visualizations? Evaluating Visualization Literacy and Reasoning Errors in Chart Interpretation** 2025

- Developed a three-phase framework to evaluate GPT-4 and Gemini 1.5 Pro on 100 misleading charts, assessing chart classification, trend interpretation, and critique using a taxonomy of 74 semantic and perceptual flaws.
- Combined accuracy, macro-F1, and Krippendorff's alpha with qualitative reasoning analysis; chart-type recognition accuracy was 81% for GPT-4 and 73% for Gemini 1.5 Pro.
- Identified recurring failures involving subtle visual distortions, unsupported assumptions, missing context, and visibility constraints.

## Publications

*Archived conference materials*

1. **Yang, C.** (2026). *Hands-on Ethical Inquiry: A Participatory and Exploratory Lab Model for AI/ML Ethics Education.* [Lightning talk slides]. 2026 NAIRR Annual Meeting, Arlington, VA, March 10–13. Zenodo. doi:10.5281/zenodo.19193449.

**Related presentations:** Lightning talk speaker, poster presenter, and live demo presenter at the same meeting (March 2026).

## Awards and Funding

**Departmental scholarship**, UMBC Fall 2026
Awarded $4,000 in graduate scholarship support.

## Service

**Program Committee Member - SIGCSE TS 2027** Oct. 2026-Present
ACM, Association for Computing Machinery

## Teaching Experience

**University of Maryland, Baltimore County**
*Teaching Assistant* Baltimore, MD

**CMSC 462: Introduction to Data Science** Fall 2026

- Support teaching and student learning in introductory data science.

**CMSC 441: Design and Analysis of Algorithms** Fall 2026

- Hold office hours, grade assignments, and help proctor exams.

**CMSC 202: Computer Science II** Fall 2025

- Led three weekly practice sessions with approximately 22 students each, serving more than 60 students in total.
- Conducted course-wide exam review sessions for a course enrolling more than 300 students.
- Held office hours, graded programming assignments, and assisted with exam proctoring.

## Professional Experience

**University of Maryland, Baltimore County** September 2025–Present
*Graduate Assistant* Baltimore, MD

**Ruijie Networks** July 2023–September 2025
*Machine Learning Engineer* Shanghai, China

- Led an algorithm team that built and deployed a fine-grained endpoint identification module for enterprise network-management products.
- Applied machine learning to mirrored network traffic and endpoint features to identify device models, types, and operating systems, including specialized medical and financial equipment.
- Supported asset inventories and device-based access policies in customer environments with hundreds of thousands of endpoints, with deployments across major state-owned enterprises, financial institutions, universities, and hospitals.
- Built intranet proxy-detection algorithms deployed at multiple universities.

## Skills and Languages

**Programming:** Python, Java, C++
**Tools:** scikit-learn, Apache Spark, Blender, LaTeX, Excel
**Languages:** English, Mandarin